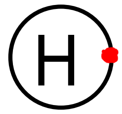
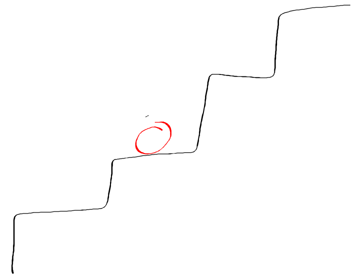
Energy Levels

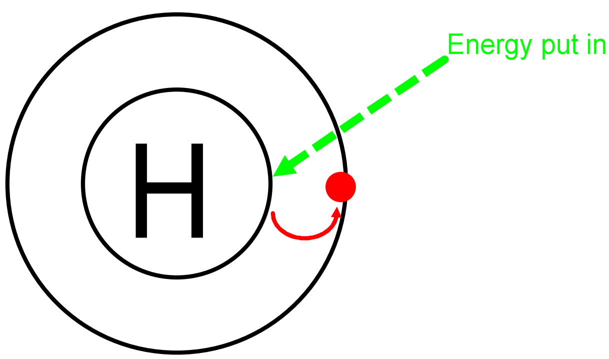
Recall the Bohr model of the hydrogen atom:





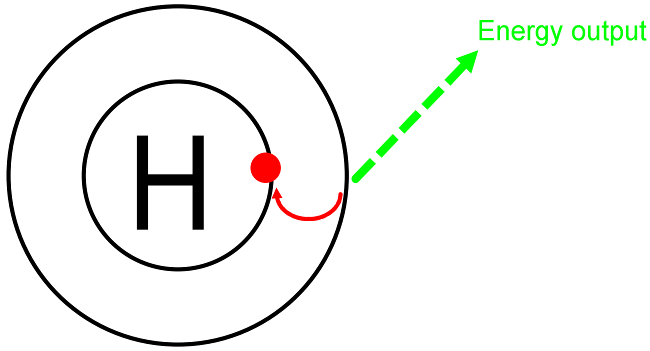
Electrons are restricted to certain orbits, or energy levels. The energy of electrons is **quantized**. i.e.. the possess <u>*specific amounts*</u> of energy at each level.

When energy is added to atom, this adds energy to the electrons, which causes them to jump up to the next energy level, or orbital.



An electron that occupies a higher energy level is said to be in an **excited state**.

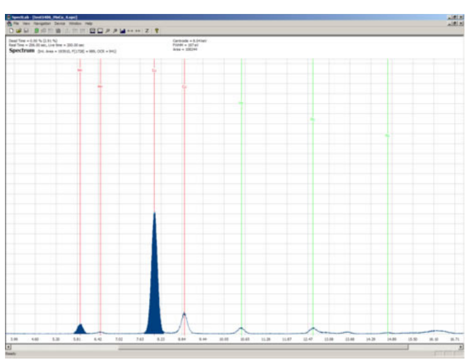
Once energy stops being put in, the built up energy is released as the electron falls back down to its **ground state**. This energy is released in the form of light.



Since all energy levels are quantized, the energy released is also quantized. Therefore, each element will release a specific, unique, light frequency.

Using a process called **spectroscopy**, we can analyze samples to determine what compounds/elements are present and at what amounts.

While there are different kinds of spectroscopy, they all opperate on the same principles. For example:



http://www.toivel.com/images/spectroscopy-analysis-software3.jpg

Each peak represents a different compound or element.

The x-axis is the wavelength of light detected, thus identifying the compound/element.

The y-axis is a relative measure of amount.